Erdem Çiçek, Ph.D.

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Solution Architect @OBSS

Innovative and results-driven Solution Architect with over 15 years of experience in enterprise application development, cloud infrastructure design, and digital transformation projects. Currently leading cloud and MLOps architecture initiatives at OBSS. Proven expertise in multi-cloud solutions (Huawei Cloud, Azure, AWS), Kubernetes-based platforms (OpenShift, CCE, AKS), and full DevSecOps enablement across diverse industries, including finance, e-commerce, and health tech.

Extensive background in system migrations (on-prem to cloud, cloud to cloud), AI/ML platform setups, and developer productivity optimization. Adept at leading pre-sales, PoC deployments, and complex architectural transformations with a strong foundation in .NET, Python, and JavaScript technologies. Recognized speaker and contributor at major industry events such as Microsoft Azure Day, AWS Cloud Day, Red Hat Summit, Huawei Cloud Partner Event, Gitlab DevSecOps Days, and GitHub Advanced Security.

Recently earned a PhD in Computational Science and Engineering at Istanbul Technical University. My doctoral research focused on enzyme catalysis and biomolecular simulations, utilizing quantum mechanics (QM) methods, molecular dynamics (MD) simulations, and AI-supported analysis to investigate druggable targets such as PAD2 and SARS-CoV-2 M_{pro}. This interdisciplinary work resulted in multiple peer-reviewed publications and bridges advanced computational research with real-world biomedical and AI applications.

Experience

Solution Architect — July 2024 – Present
OBSS Technology — *Remote – Turkey*

- **Modanisa.com**
  - Migration of **GCP** infrastructure to **Azure** (consultancy, and delivery - ongoing)
    - Comprehensive GCP Infrastructure Assessment using Github Copilot Custom Agents, including HLD and application diagrams.
    - PoC setup for the App team to test Azure container app features.
    - Azure Well-Architected Framework with Hub-spoke network with Virtual WAN and terragrunt development using Github Copilot Custom Agents.
- **HayatFinans**
  - Ongoing Pre-Sales Activities for **MLOps Platform**: Supporting dedicated ML/GenAI workloads.
    - **OpenShift AI** demo, PoC, and HLD for cluster design.
    - Cluster provision using NMState Operator to provide dev-prod network isolation for egress, and ingress sharding to provide network isolation for ingress traffic.
    - Mixed MIG configuration, custom workbenches, custom runtimes, custom model inference.
- **CoinPara**
  - Migration of on-prem workloads to the **Huawei Cloud** (consultancy and operation - ongoing)
    - PoC for **CCE AutoPilot** and RDS services.
    - Hub-spoke network for dev and prod environments.
    - Solution and migration of CCE AutoPilot workloads, hub resources, and DB and middleware resources.
    - terrgrunt development using Github Copilot Custom Agents.
- **Getir Araç**
  - Migration of **AWS** infrastructure to **Huawei Cloud** (consultancy)
    - Comprehensive AWS Infrastructure Assessment, including HLD and application diagrams.
    - PoC setup for the DB team to test ELB features.
    - Hybrid-Cloud architecture design with network layer and Huawei Cloud on-cloud solution
- **Aselsan**
  - Ongoing Pre-Sales Activities for **MLOps Platform**: Supporting dedicated ML/GenAI workloads.
    - **OpenShift AI** demo, PoC, and HLD for cluster design.
    - Custom solution using **Vanilla k8s, Kubeflow, KubeRay and AIBrix**
- **TEB**

    * Consultancy for **Openshift AI**
        · provided solution for **GPU virtualization** and **MIG** profile setup
        · Provided technical documents for deployment of Mistral7B LLM using RHOAI
        · Phase 2: deployment of LLMs and RAGs (ongoing)
    * Consultancy for **Azure DevOps Server 2020** (ongoing).
        · Provided several pipelines and integration solutions for the DevOps team.
    * PoC for **AWS** EC2 UltraCluster (ongoing).
        · Provided a solution for **model fine-tuning** and **model training** jobs using EC2 UltraClusters (ongoing).
- **Şekerbank**
    * PoC and consultancy for migration of **SAS Viya** ML workloads to the **Openshift AI**
        · PoC : Help the infra team to prepare the RHOAI environment on current dev OpenShift clusters.
        · PoC : Provided custom workbook image templates that include required libraries for ML development. Also provided custom runtimes for model inference.
        · Phase 2: Consultancy for full migration from SAS Viya
- **Hayat Finans**
    * Pre-sales activity for **OpenShift AI** to set up an end-to-end MLOps platform for a dedicated data science team.
- **VakıfBank**
    * Pre-sales activity for **OpenShift AI** to set up an end-to-end MLOps platform for a dedicated data science team.
    * Delivered a research report that is providing detailed information about several MLOps platforms and tools.
- **Paymix**
    * Consultancy and operation for **OpenShift** to set up an end-to-end Kubernetes environment for core banking app deployment.
    * Health check and troubleshooting on the dev cluster using must-gather logs.
- **Scoutium**
    * Migration of **AWS** infrastructure to **Azure** (consultancy and operation)
        · Comprehensive AWS Infrastructure Assessment, including HLD and application diagrams.
        · Azure Well-Architected Framework for EKS to AKS migration.
        · Hub-spoke network, AKS with AGIC and Application Gateway.
- **Fintag**
    * **Azure** Subscription Move (consultancy and operation)
        · Provided consultancy for migration of all resources to the partner subscription.
        · Azure Container Apps configurations for app deployment.
    * Migration of on-prem workloads to the **Huawei Cloud** (consultancy and operation)
        · PoC for **CCE AutoPilot** and RDS services.
        · Hub-spoke network for dev and prod environments with Azure AppGW integration.
        · Solution and migration of CCE AutoPilot workloads, hub resources, and DB and middleware resources.
        · setup of 3 node ELK cluster on ECS
- **Heltia**
    * **Azure** Subscription Move and infra optimization to provide private network (consultancy and operation)
        · Azure Well-Architected Framework with Hub-spoke network and Terraform
        · Private Azure App services with Azure App GW.
    * Preparation of Azure audit document for partnership.
- **DopingHafıza**
    * Migration of **AWS** infrastructure to **Huawei Cloud** for providing a BCDR site. (consultancy and operation)
        · Comprehensive AWS Infrastructure Assessment, including HLD and application diagrams.
        · PoC for API Gateway custom authentication with Keycloak. Provided a custom Node.js solution for Keycloak-API Gateway integration.
        · Set up an S2S VPN connection between AWS and Huawei Cloud VPCs.
        · Tested DRS for PostgreSQL migration and SMS for Keycloak EC2 migration.
- **Agesa**
    * PoC for **GitLab** to set up an end-to-end DevSecOps platform for DevOps and security teams.
        · Pipeline migration from Jenkins for 2 Java repos.
        · SAST and GitLab Duo tests

- **Sefine Shipyard**
  - Setup and consultancy for **GitLab** to set up an end-to-end DevOps platform for developer teams.
    - Preparation of several pipelines for python based application with runner setup.
    - provided solution and implementation for backup/restore and monitoring.
- **SunExpress**
  - Developer Productivity Workshop
    - DORA, Space, and DevEx details.
    - Provided an assessment report based on the interviews and surveys with all team members.
- **Other Activities**
  - Developer Productivity
    - Pre-Sales activities for **Hatica, Valven and DevEx** for Garanti Bank, ING Bank, Defacto, Aselsan
  - SDLC Transformation and Optimizations
    - AI/ML tools benchmark for frontend and backend development.
    - ABB Bank (Azerbaijan): Consultancy for improving efficiency using AI tools for SDLC.
    - **Cobol** to Java : legacy code transformation using LLM LLM Agents with **CrewAI**
    - .net framework to .Net : legacy code transformation using LLM Agents with **CrewAI**
    - Agentic AI solutions in SDLC
    - **STM** : Migration of legacy code repositories from IBM ClearCase to BitBucket (git) using python
  - Events
    - **Azure specializations** proof preparation and audit session for: Kubernetes on Microsoft Azure, Infra and Database Migration to Microsoft Azure and Agentic DevOps with Microsoft Azure and GitHub
    - **Empower Today, Build the Future: Azure (2025)** : Heltia Azure Well Architected Framework Solution. (presenter).
    - **Powering the Future of DevSecOps with Gitlab (2025)** : Gitlab Duo and AWS Q Dev Demo (presenter).
    - **AWS Cloud Day 2024** : Amazon Q Builder, Features, and Benchmark Results (presenter).
    - **Red Hat Summit: Connect 2024** : How we implemented OpenShift AI for TEB. (content provider)
    - **Github Advanced Security 2025** : How Github Advanced Security helps developer teams to secure their codes. (presenter)
    - **Huawei Cloud Partner Elite Camp 2024** : Selected as elite pre-sales member for 2024.

Cloud Solution Architect (Team Lead) Sep 2023 – Jul 2024
Huawei R&D Center *Full Time – Turkey*

- **Defacto Cloud Migration (Oct 2023 – July 2024)**
  - Lead Solution Architect for the migration of on-premises infrastructure to **Huawei Cloud**.
  - **Phase 1 (Completed):** Migration of an upgraded .NET e-commerce application to Cloud Container Engine (CCE):
    - Joined as Lead SA post-deployment to support SRE teams in resolving latency and pod auto-start issues.
    - Introduced profiling tools to analyze **Thread Pool usage**, **Heap Memory Objects**, and **Garbage Collection Events** in production.
    - Collaborated with customer development teams to fix improper async implementations and optimize memory usage.
  - **Phase 2:** Migration of internal applications (700+ VMs, 2 OpenShift clusters) to Huawei Cloud:
    - Led pre-sales technical support and produced **high-level solution designs**.
    - Completed the **Resource Discovery and Assessment (RDA)** process and proposed optimized resource configurations with quotations.
    - Designed and delivered two **PoC plans** for migrating OpenShift workloads to **CCE** and **MSSQL RDS**; provided technical support during execution.
    - Authored **High-Level Design (HLD)** documents for network architecture and account management strategy.
    - Defined **migration roadmaps** and created a **feature request** to HQ for inter-VPC traffic control via firewall.
- **Hepsiburada Cloud Migration (Feb 2024 – July 2024)**
  - Cloud Solution Architect for the migration of ELK clusters runinin in GKE to Huawei Cloud CCE Turbo.
  - Contributed to the PoC design for ELK Clusters using **CCE Turbo**.
  - Network design for Hybrid Cloud environment, Direct Connect and Enterprise Router setup for On-Prem-Huawei Cloud Zone 1 connection.
- **Logo Software Cloud Migration (Oct 2023 – Completed)**
  - Cloud Solution Architect for the migration of a multi-tenant SaaS application from on-prem Kubernetes to **CCE**.

    * Delivered **HLD** for CCE-based deployment and implemented **Ansible** and **Terraform** automation for provisioning.
- **TIM – CloudBlue – Huawei Cloud Integration (Jan 2024 – July 2024)**
    * Cloud Solution Architect for integrating Huawei Cloud core services into the CloudBlue marketplace platform.
    * Provided technical guidance on the **APS development framework** and Huawei Cloud **REST API layer**.
- **CodeArts DevOps Services SA (Oct 2023 – July 2024)**
    * Designed various use cases for advanced DevOps scenarios.
    * Submitted enhancement requests for new feature support to Huawei HQ.

Solution Architect Jan 2023 – Sep 2023
Rise Technology, Consulting & Academy *Remote – Turkey*

- **Hayat Holding**
    * Provided consultancy for the migration of on-premises infrastructure to the **Azure Cloud**.
    * Served as **Team Lead** for two enterprise web portal projects:
        · Designed technical solutions for the new **SOA architecture** with SAP integration and implemented **Azure Service Bus**.
        · Developed an n-tier **.NET 7 – Telerik Kendo** web application and **REST API** using best practices such as DDD, TDD, CQRS, and Vertical Slice architecture.
        · Created **proof-of-concepts (PoCs)** for various **Azure** services.
        · Provided consultancy on project management and task planning using **Jira Cloud**.
- **Borusan Holding**
    * Assessed the existing **SOA architecture** and back-end services against modern best practices.
    * Reviewed and evaluated the code base of .NET applications.
    * Proposed a cloud-native **microservice architecture** using **Dapr** on **Azure Kubernetes Service (AKS)** to address enterprise-level needs.
- **Otokoç**
    * Led the cloud-native modernization of an ERP transformation project as a **Full-Stack Developer**:
        · Designed and implemented a microservices architecture.
        · Utilized Docker, AKS, .NET Core, Redis, ELK stack, and RabbitMQ.
        · Refactored backend code to resolve performance bottlenecks.
        · Developed the admin panel frontend using **DevExtreme jQuery**.
    * Contributed to the modernization of the Customer Portal (Otokoç 2. El) as a **Full-Stack Developer**:
        · Designed microservice-based backend components.
        · Developed admin panel frontend using **DevExtreme jQuery**.
        · Implemented the authentication and authorization layer using **IdentityServer4**.
- **Defacto**
    * Provided technical consultancy for resolving **ELK stack** integration issues with **Azure DevOps Server**.
    * Delivered solutions for a **PostgreSQL cluster** implemented with **Patroni** and **HAProxy**.
- **Microsoft Azure Advanced Specialization**
    * Contributed to the organization's efforts to attain **Azure Advanced Specialization** certification.
- Delivered introductory **C# training** for the **"yenidenbiz.org"** initiative.
- Provided mentorship to junior, mid-level, and senior developers, supporting career planning and internal communication within the organization.

Lecturer / Researcher / Software Developer Mar 2018 – Jan 2023
Istanbul Technical University *Istanbul, Turkey*

- Developed components using **SharePoint 2019** and **Nintex Workflow** for SharePoint 2019 with CSOM.
- Implemented **.NET Core 3.1 Web API**, Entity Framework Core, and CQRS with MediatR; wrote unit and integration tests using NUnit and Moq; contributed to frontend development with Vue.js 2.6 and mockups.
- Integrated **Azure Translation Service** for real-time content translation.
- Worked with DevOps tools such as Jenkins, Jira, Confluence, and Bitbucket.
- Used **Serilog** for logging to an **ELK stack** running in Docker containers; monitored with **Kibana** and managed containers via **Portainer.io**.
- Developed efficient and maintainable software aligned with the objectives and needs of academic staff and students (graduate and undergraduate level).

- Designed and implemented databases and table structures for web applications.
- Technologies used: C#, ASP.NET, SharePoint 2010, Nintex for SP2010, Telerik ASP.NET Toolkit, SQL Server 2008.
- Conducted computer-aided drug design studies and theoretical research on biomolecular reaction mechanisms.
- Served as a teaching assistant for programming courses.

Visiting Researcher — Apr 2018 – Jul 2018
Université de Lorraine — *Lorraine, France*

- Performed multiple molecular dynamics simulations of enzyme–ligand complex systems.
- Tools: Amber18, cpptraj, **Python, Matplotlib, Pandas, NumPy**.
- Prepared various analyses, reports, and documentation using **Markdown** and **LaTeX**.

Research Assistant / Software Developer — Mar 2009 – Mar 2018
Istanbul Technical University — *Istanbul, Turkey*

- Developed efficient and maintainable software aligned with business objectives and the needs of academic staff and students.
- Designed and implemented databases and table structures for web applications.
- Developed components using SharePoint Framework, integrated third-party services, and utilized CSOM.
- Technologies used: C#, ASP.NET, SharePoint 2010, Nintex for SP2010, Telerik ASP.NET Toolkit, SQL Server 2008, XML Web Services (Scopus and Web of Science).
- Conducted computer-aided drug design studies and theoretical investigations of biomolecular reaction mechanisms.

Junior Developer — Jul 2008 – Mar 2009
Broadage Software Company — *Istanbul, Turkey*

- Developed desktop software using C#.
- Implemented web services to retrieve data from various live sports competition sources.
- Designed databases for horse racing competitions.
- Collaborated with team members to resolve issues and apply best practices.

Education

Istanbul Technical University — Istanbul, Turkey
PhD. in Computational Science and Engineering — *Jul 2019 – Jul 2025*

Istanbul Technical University — Istanbul, Turkey
MSc. in Computational Science and Engineering — *Sep 2011 – Jul 2014)*

Istanbul Technical University — Istanbul, Turkey
MSc.(non-thesis) in Bioinformatics — *Sep 2008 – Jul 2010)*

Istanbul Technical University — Istanbul, Turkey
BSc in Molecular Biology and Genetics-(Final Project completed with the dept. of Computer Engineering) *Sep 2003 – Jul 2008)*

Publications

- **Çiçek, E.**, Munar, İ., Çınar, S. A., Başçeken, S., Monard, G., Aviyente, V., Sungur, F. A. (2025). Computational Insights into Substrate-Assisted Citrullination Mechanisms of PAD2 Isozyme: A Comparative Analysis of Reaction Pathways. Journal of Molecular Graphics and Modelling, 109107.
- **Cicek E**, Monard G, Sungur FA. Molecular Mechanism of Protein Arginine Deiminase 2: A Study Involving Multiple Microsecond Long Molecular Dynamics Simulations. Biochemistry. 2022;61(13): 1286-1297
- **Cicek E.**, Sungur FA. Understanding the Mechanism of PAD2 Using Multiple Microsecond Long Molecular Dynamics Simulations. 13. The International Symposium on Health Informatics and Bioinformatics HIBIT, October 27-23 2020, Istanbul, Turkey.
- Yuce M, **Cicek E**, Inan T, Dag AB, Kurkcuoglu O, Sungur FA. Repurposing of FDA-approved drugs against active site and potential allosteric drug-binding sites of COVID-19 main protease. Proteins. 2021;89(11): 1425-1441.
- **E. Cicek**, O.Salman, "Blood Flow Analysis of Aneurysm Data Using ParaView Co-Processing Tools" International Conference on Mechanical Engineering Design ICMED 2016, 25-26 April 2016, Chennai, India.

- **E. Cicek**, A. Konuklar, “Elucidating the Transamination Mechanism-Based Inactivation of Dopa Decarboxylase by Using Computational Methodologies” Theory and Applications of Computational Chemistry TACC 2012, 2-7 September 2012, Pavia, Italy.
- **E. Cicek**, Quartz Binding Peptide Design Using Alignment Scores, Physiochemical Properties and Feature Selection Techniques, Istanbul Technical University, 2008

Certifications and Courses

- HCIA & HCIP - Huawei Certified ICT Professional
- Software Specialist (C#) - Bilge Adam Academy, 2006 (400 hours)
- Database Management - Bilge Adam Academy, 2006 (100 hours)
- Web Development with ASP.NET and SQL Server - Istanbul Technical University IT Office, 2009 (50 hours)
- Molecular Dynamic Simulations of Proteins: Fundamentals and Applications, TOBB Economy and Technology University, 2015 (75 hours)
- Bioinformatics Summer School, 2014 (60 hours)

Hobbies

- Sport (Football, Running)
- Loves to meet new cultures and peoples
- PC Games
- Loves to test new tools on my home server
- Continuous learning and research

References

- Batuhan Aydemir (Technical Account Manager)
  - Vodafone
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NOTE: Additional references may be furnished upon request.